

Data Modeling

OLAP vs OLTP

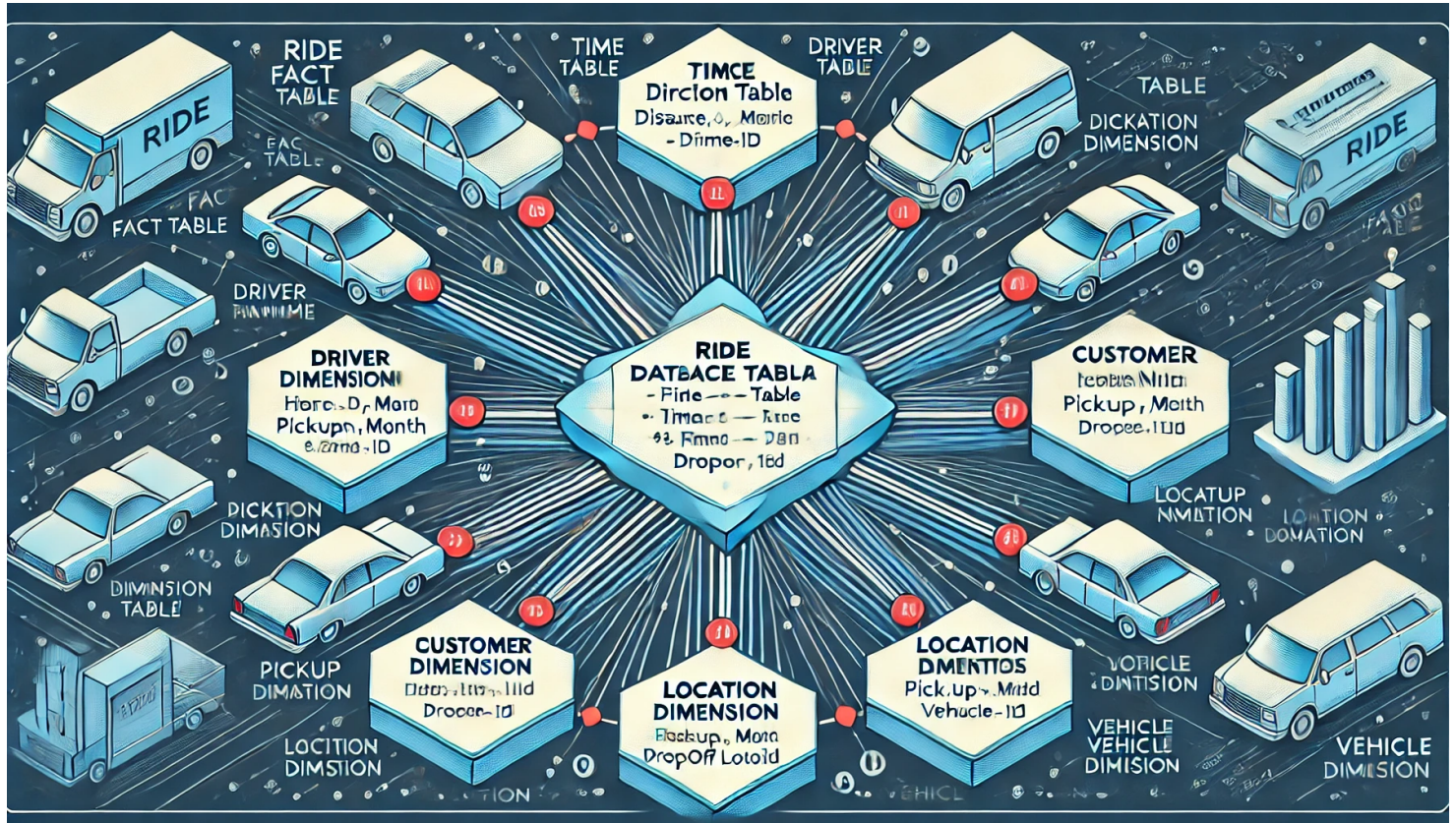
Introduction to Data warehousing ✓

What is Data Warehouse ✓

How to design a Data Warehouse

Dimensional Modeling Approach: star ,galaxy and snowflake schema

Slowly Changing Dimensions





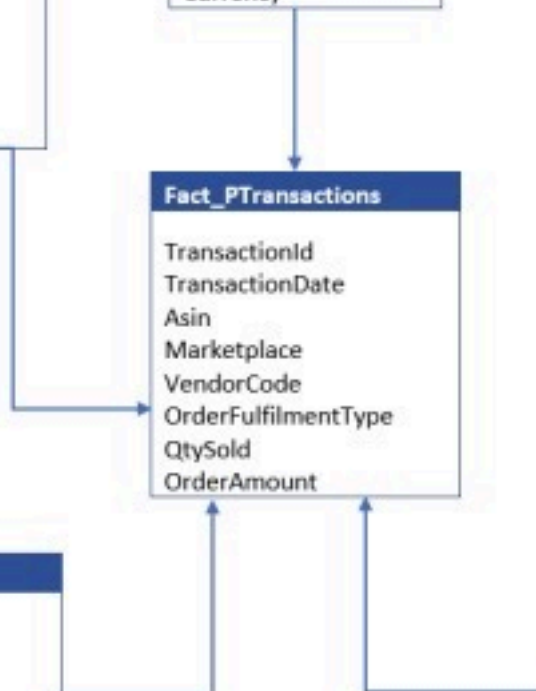
Dim_PrintBooks
PK_Asin
BookName
PrimaryAuthor
SecondaryAuthor
Language
PaperBack
ItemWeight
UnitPrice

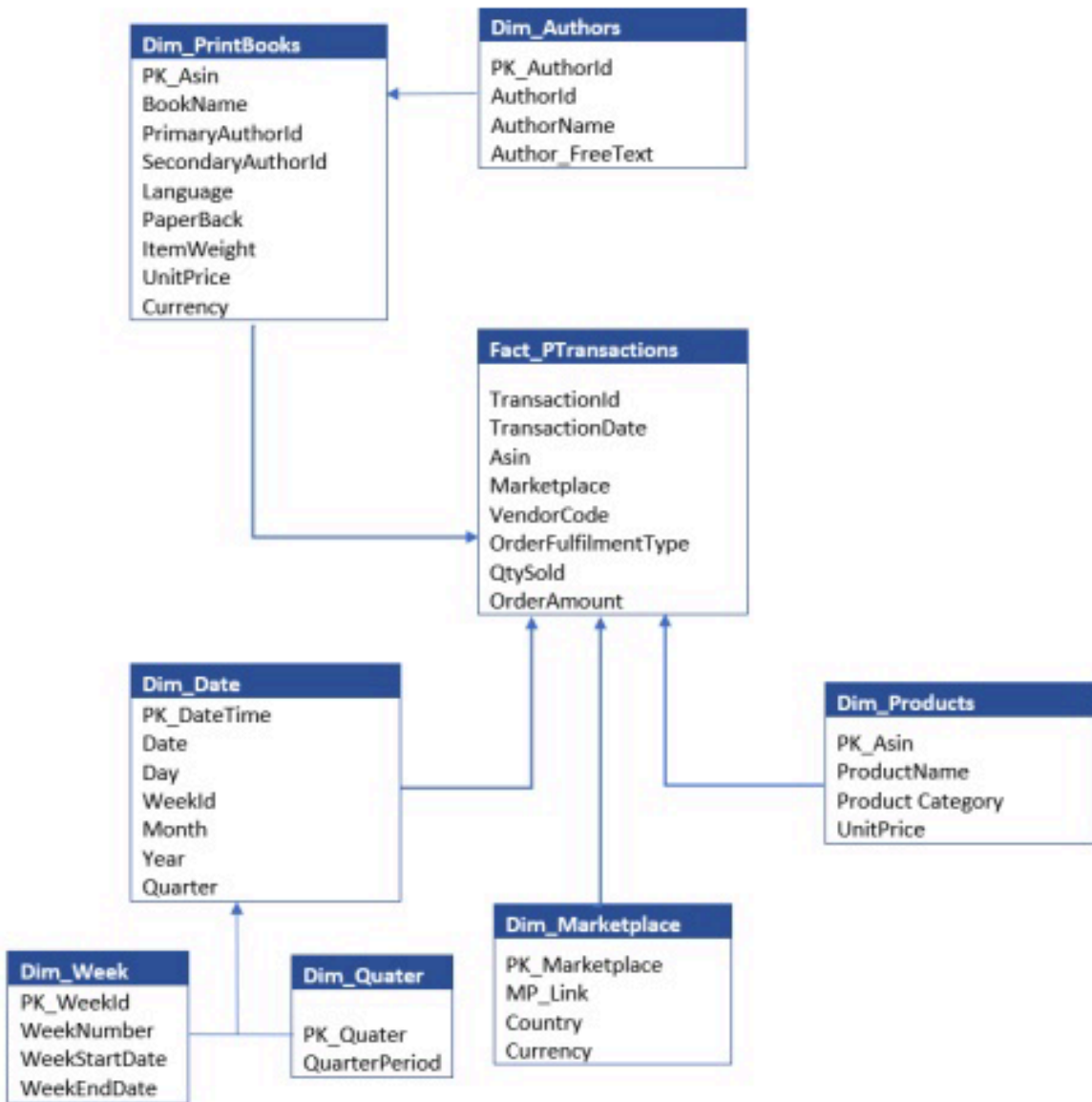
Dim_Marketplace
PK_Marketplace
MP_Link
Country
Currency

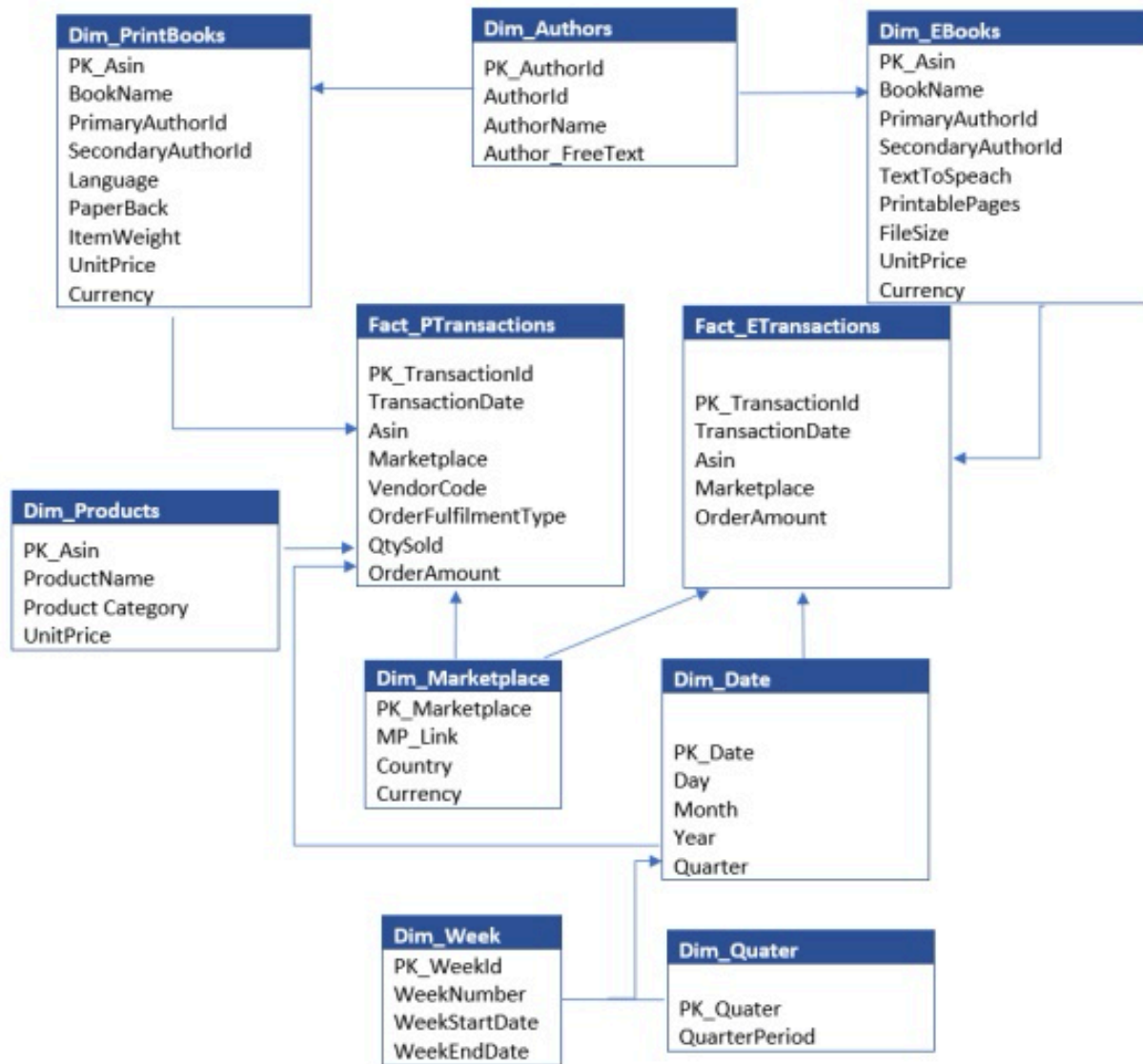
Fact_PTransactions
TransactionId
TransactionDate
Asin
Marketplace
VendorCode
OrderFulfilmentType
QtySold
OrderAmount

Dim_Date
PK_DateTime
Date
Month
Year
WeekNo
WeekStartDate
WeekEndDate
Quarter
QuarterPeriod

Dim_Products
PK_Asin
ProductName
Product Category
UnitPrice

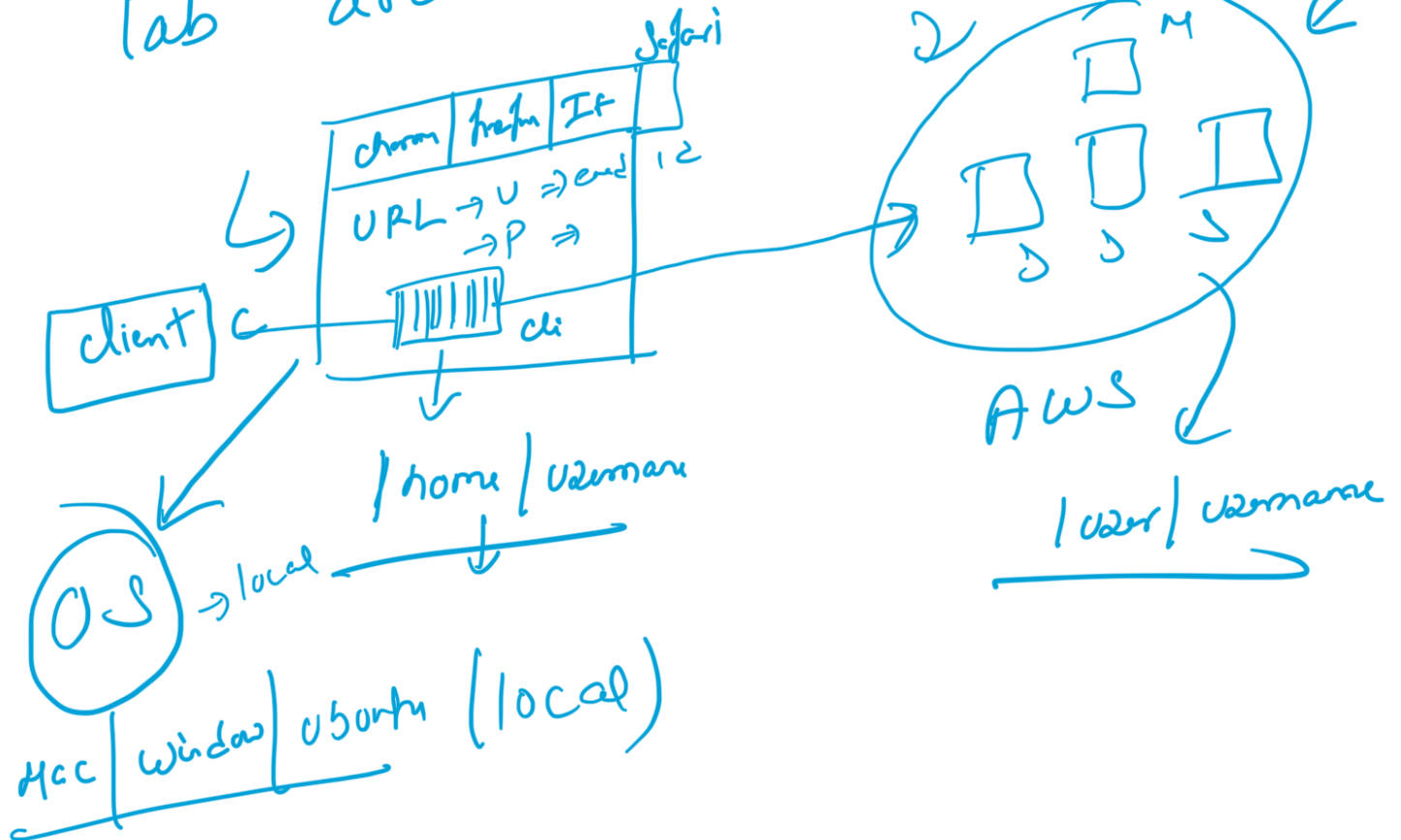




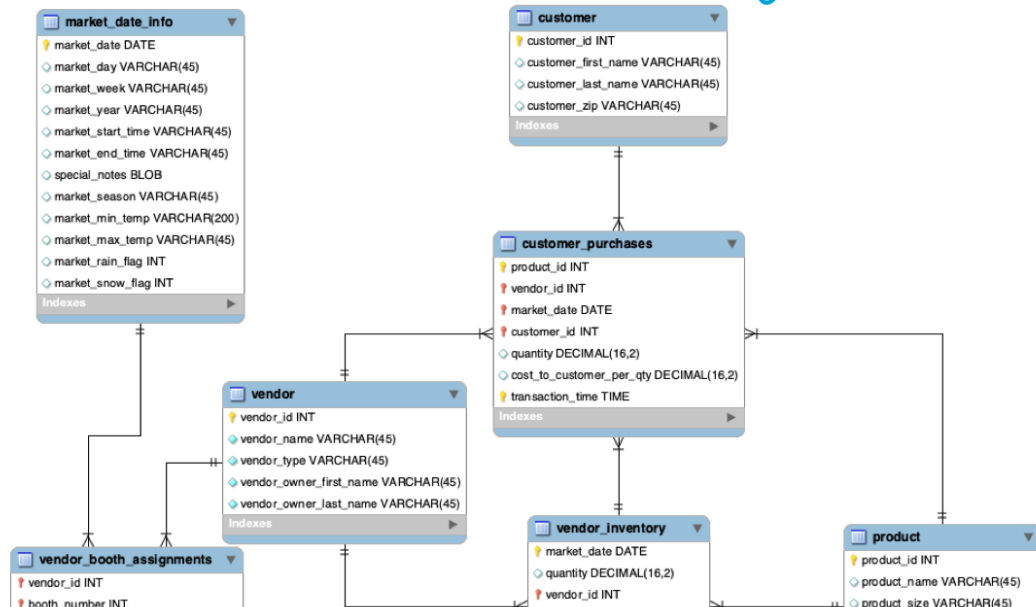


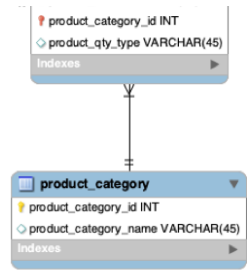
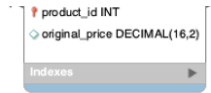
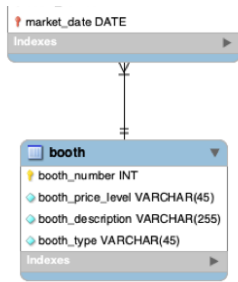
class begins of 9:05PM
(GUI)

lab architecture



Data Modelling





OLA P System $\Rightarrow$ Data warehouses

- $\Rightarrow$ Reads
- 4 Analytical queries to derive patterns.
- $\hookrightarrow$ structured (Tables)
- $\hookrightarrow$ Denormalized

Dimensional Modelling

① Fact table : where the action happens.

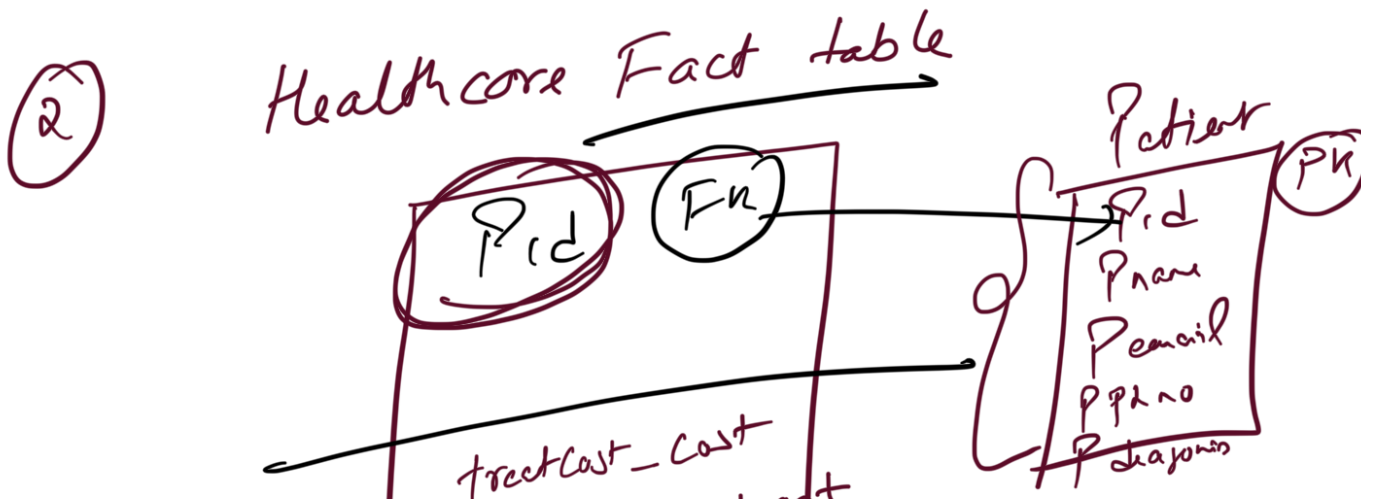
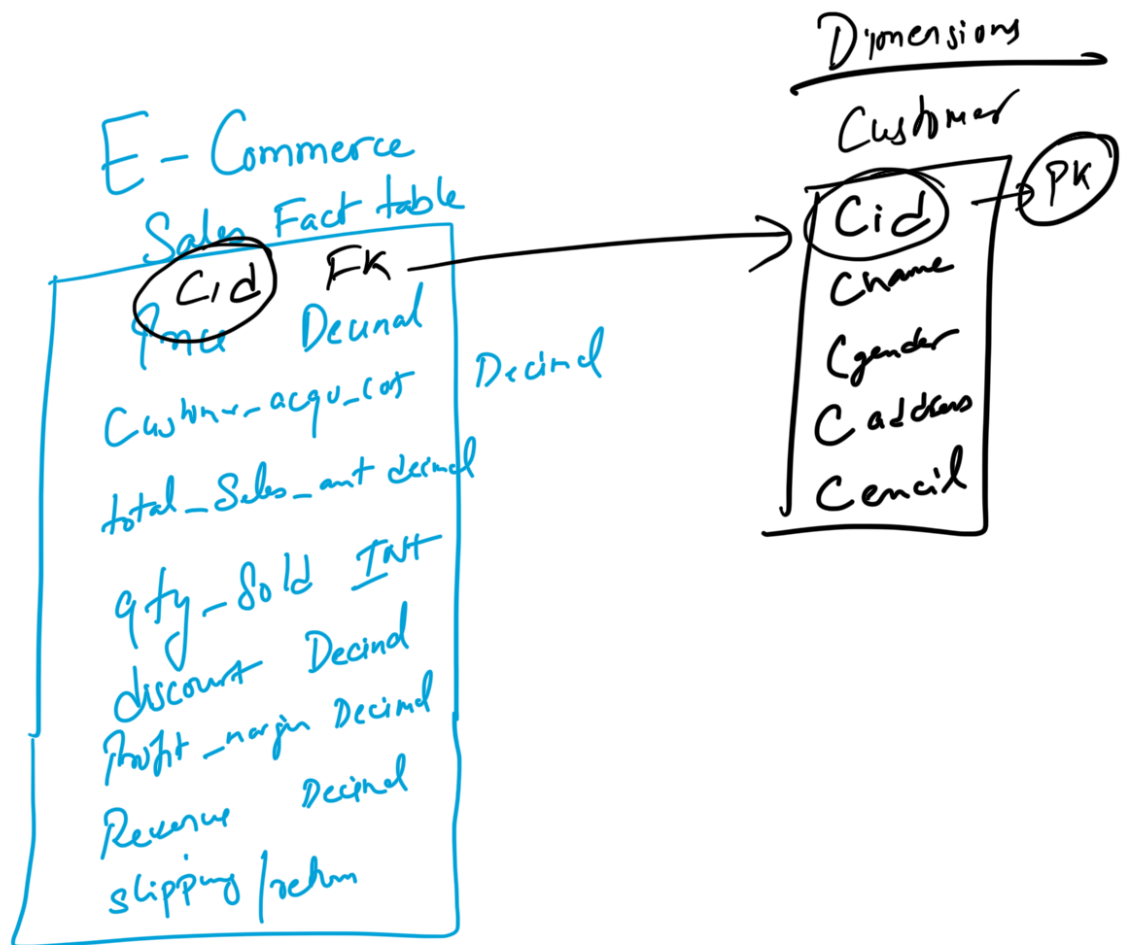
② Dimension table

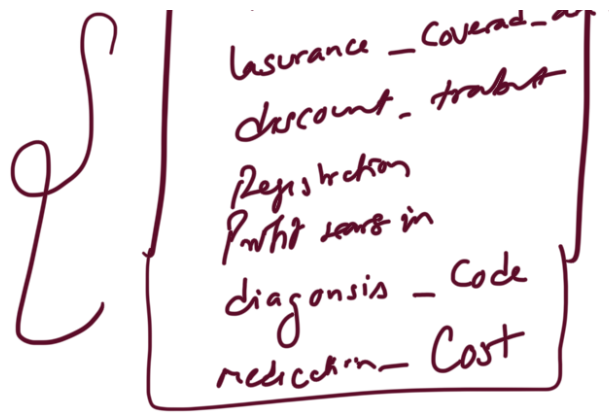
① Fact table :- It is the place where all your core data is stored.

② Dimension table :- It is the place where all your reference data is stored.

① Business measure
Can be measured.

↳ Contain numbers. ②
↳ They do contain FK from other tables.

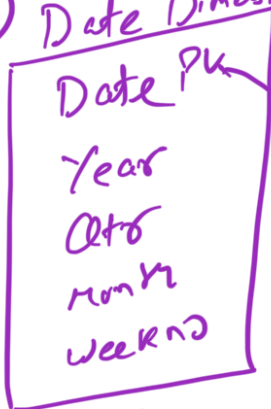




- (2) Dimension tables:- they provide context to your data model.
- ↳ that helps you make sense of raw data in fact table.
 - ↳ they contain descriptive attributes (dimensions)

(1) E Commerce

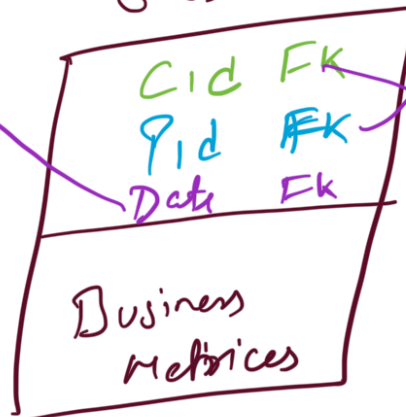
* Date Dimension



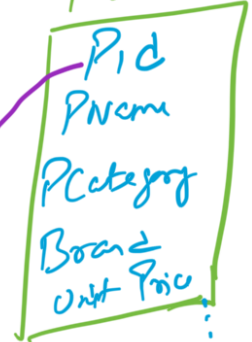
timestamp

Day of week
is holiday

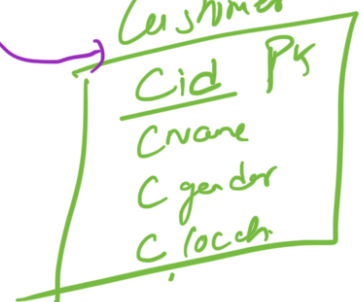
SalesFact



Product



Customer



Schema Design (80)

① Star Schema

- ↳ Simple and intuitive choice
- ↳ only 1 Fact table
- ↳ Can have multiple Dimension table.

Why to use a star schema

- ✓ ↳ Easy to understand
- ✓ ↳ Fast Performance

⇒ ⑨ ↳ Some Redundancy

15%

②

Snowflake Schema: - when you need more normalization

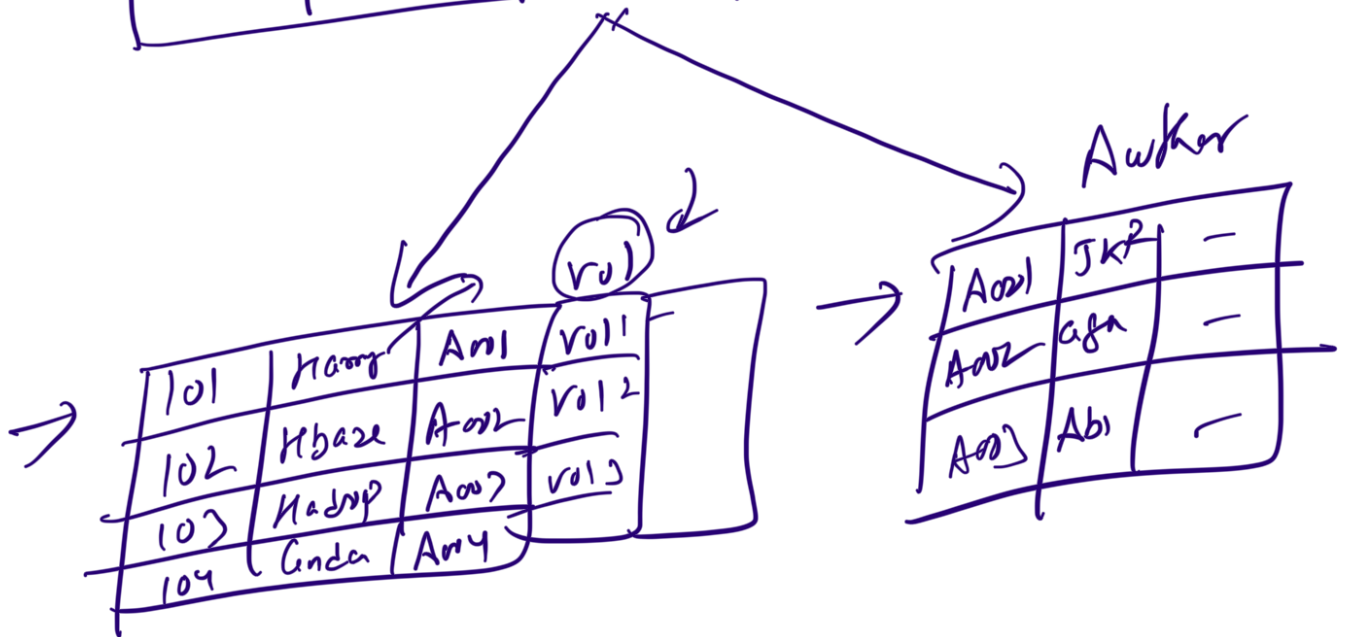
⇒ 1 Fact table

⇒ Multi level Dimension tables

Star schema

101	Harry Potter	J.K Rowling	1998
	Harry Potter	"	1999

102	1)	1)	2000
103	1)	1)	2001
104	1)	1)	



Snowflake :-

↳ Save space

↳ more joins

(3)

Galaxy Schema / fact constellation

5%

↳ Multiple fact tables

↳ Shared Dimension tables

